Multi-scale and Multi-domain Edge Determination with Accurate Gradient Orientation Computation in Aquatic Environment

Luke Bagan
Department of Mechanical Engineering
University of South Carolina
Columbia, SC, USA
lbagan@email.sc.edu

Yi Wang
Department of Mechanical Engineering
University of South Carolina
Columbia, SC, USA
yiwang@cec.sc.edu

***Abstract*— New methods for accurate edge determination and gradient orientation computation are proposed. When analyzing local gradient maxima, the error in gradient angle can propagate to deteriorate edge quality. The proposed method examines each individual pixel and considers pixel-wise gradient magnitude along various orientations. This gradient orientation computation method synergizes with a new edge determination method based on multiple domains and scales. This method uses enhanced non-maximum suppression paired with a Gaussian Mixture Model (GMM) to fuse a scale space of gradients spanning multiple image modalities.**

Keywords—Image Processing, Gradient Computation, 2D Filtering, Edge Detection, Scale Space, Noise, Cubic Spline Interpolation, Gradient Orientation,

I. Introduction

Edge detection and its associated properties are some of the most important steps in image processing pipelines as they reduce and enrich the information within an image for minimal complexity and establish a foundation for feature-based processing techniques. Accurate edge detection must consider a variety of factors including noise, image sharpness, orientation, empirical parameters, and computational complexity. Like many processes today, edge detection has become dominated by deep learning (DL) models. The performance of DL methods, however, is restricted to the data it was trained on, which is subject to data scarcity [1]. Robust low level image analysis methods eliminate this limitation and are well suited to address common obstacles with the implementation of a rigorous multi-scale, multi-domain edge determination methods.

One critical step in edge detection is computing gradient orientations. Generating superior pixel orientations not only allows for deeper analysis of aquatic images, but accurate gradient angles also allow edges to be tracked, grouped, and linked according to their orientation for analysis on waves, surface currents, or objects. Essentially orientation offers another feature point for deeper analysis of the image contents. [2] Enhanced gradient angles also set up the foundations for edge determination methods that rely on the orientation to improve overall quality.

This work combines novel gradient computation filters with multi-scale, multi-domain gradient computation for edge detection. It first enables smooth edge detection without discontinuities at problematic areas like corners and curves by increasing the fidelity of non-maximum suppression (NMS) with better angles. It then examines the edge probability at various angles using a probability density function decomposition. This is heavily inspired by the human vision system, which analyzes a scene at multiple scales to ensure it can detect features of all sizes [3]. Our detection functions are all performed on a multi-scale combination of gradients to help minimize line breakages, suppress noise, and offer additional gradient information enabled by more image domains.

II. Related Works

Many previous research efforts on edge detection have been reported in the area of image processing and computer vision. Traditional methods are highlighted by gradient and derivative filters. The most common of which are Prewitt and Roberts, Sobel [4], LoG [5], and Canny [6]. Each of these methods uses some kernel, typically 3x3, with a specific pattern and convolves around an image to produce the intensity derivatives of each pixel [7]. Prewitt and Roberts, and Sobel are built on central difference, and measure the difference in intensity at a pixel using the horizontal and vertical neighbors. However, Sobel varies from the other methods by weighing the pixels closer to the center.

All these traditional methods are quite sensitive to noise. So, some research has gone into deploying larger kernels with variations in sizes 5x5, 7x7, and more to suppress noise in images and improve the resultant features [8]. The Canny method was developed as another method to reduce the impact of noise because it uses double thresholds and hysteresis which only allows edges to be defined at or near a strong gradient.

While easy to apply to an image, the accuracy of the traditional methods heavily depend on image quality. Most gradient-based operators have a simplistic structure, which limits their performance. Coupled with their high pass nature, noise tends to be amplified [9] [10]. Smoothing filters are used to combat this problem but end up blurring the image, which

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can severely degrade the quality of small or detailed objects. In many applications, where the capture of small objects is important, smoothing filters cannot be used.

While gradient-based methods dominated early traditional methods, alternative methods were explored to address some of their shortcomings. The other traditional edge detection methods delve into second derivative zero crossings, region segmentation, image morphology, fuzzy logic, spatial frequency, machine learning, and more [11]. The Laplacian of Gaussian [5] (LoG) filters are built off the principle of using the second derivative at a pixel and locating an edge at the zero-crossing of that pixel. The kernel used for this is derived from the Gaussian kernel and employs a weighting scheme, which places more emphasis towards the central pixel. It was proposed as an alternative to filters based on single differentiation because it could handle orientation better. However, it displays extreme sensitivity to noise, and using a Gaussian smoothing filter, inherently reduces image quality [12].

As researchers were developing new methods of edge detection, region based [12] methods as Khan proposed [13] grew in popularity. These excelled at capturing the structure of the image and as such would produce great high-level edges but had no texture. [14] Frequency domain analysis was one of the last steps before deep neural networks started to dominate research. Edges and features would be located at maximum phase congruency, and this was considered one of the best methods of the time. [14] Like Canny proposed a framework to obtain edges from gradients, Fuzzy logic edge detectors did the same. They were developed to reduce complexity by introducing mathematical and logical reasoning. However, since Fuzzy logic still requires gradients, it essentially ran into the same issues as many of the other detectors. Noise was a big problem and could only be addressed with a high-dimensional ruleset. [15] A popular obstacle for these methods was noise and as such Lin proposed a new high-pass filter [10] for determining edges in classically noisy medical images. In fairness, it does perform well in noise, though it still does lack angular sensitivity, and low contrast/degraded boundaries.

No traditional methods have been able to truly tackle all these issues. As machine learning began to dominate other industries, they also showed promise for detecting edges. [16] DexiNed [17], BDCN [18], and EDTER [19], and TEED [20] are four popular edge detection models that top the scales in measurable performance. However, like traditional methods, machine learning methods are susceptible to some issues. The biggest problem well recognized is the demanding need for data. This is especially true for edge detection where good data is hard to come by because of the challenge in labeling data. Not only does this mean there is not a lot of data available for edge detection models, but it also means everything is trained on similar and limited data, posing a real problem for model generalization. Many models will struggle to interpret situations that they were not specifically trained on. [1]

This may be more of a problem for segmentation models and object detection models that require rich feature information present in their data. In niche settings like marine environments, datasets are extremely scarce. On top of this, the environment can be highly cluttered and extremely variable. Without human characteristics like common sense, intuition, explanation, and the ability to process something unseen, deep learning models will continue to struggle to match the performance of the human vision system.

A massive benefit of traditional methods is that they do not require data and as such are not subject to the abovementioned constraints. This means their performance should be stable across whatever data they are applied to. Moreover, the advancements in computing power of hardware are enabling these systems to mitigate their computational burden on deployed systems. This prompts the need for a dataless method that directly operates on the original image's raw pixels to determine edges from high quality gradients.

III. Overview Of Solution

Edge detection is a long studied and very popular area of research in computer vision because of its value and its difficulty. No one solution exists which has solved the issue of detecting accurate, smooth, and continuous edges because images are complex domains that pose many challenges in accomplishing this. Noise, inconsistent lighting, shadows, textures, obstructions, reflections, and more prevent smooth edges from being detected. One of the significant problems has to do with small objects in an image, which are easily degraded by noise. Therefore, they are hard to detect by traditional operators or learning based models because they do not have enough feature presence to generate any response.

In addition, the definition of an edge tends to be somewhat arbitrary and subjective. One person may consider the boundary between the shadow and the light in an image an edge boundary where another person may not since it could be an unintrusive lighting change, or even an unimportant area of the image. This poses the problem of generating quality ground truth data for edge detection models, as well as having baseline data to compare against and evaluate edge detection frameworks.

To solve these issues, a new edge detection framework is proposed, which starts from the foundation of two novel gradient-based filters, the Wide View Filter (WVF) and the Line Filter (LF) [21]. WVF and LF provide exceptional angular fidelity, which lays the groundwork for a new enhanced method for calculating image gradients. Since the rotational invariance of edge detectors is critical to its accuracy [22], the WVF and LF are well suited for this problem. Because the WVF and LF can be rotated to any desired angle, they can be run to generate a set of gradients, which covers a discrete domain from 0-180 degrees. The set of gradients can then be used to generate a piecewise cubic spline function, which can be solved to find the gradient direction corresponding to the maximum gradient. This is where the edge of a pixel lies. The new method for computing gradient orientation provides exceptional accuracy, which can then be used to enhance techniques in edge detection like non maximum suppression and yield much smoother connected lines. The gradients are also useful for several image processing techniques.

After calculating gradient angles, the process of edge detection can begin from a superior foundation. However,

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enhancing gradients is not all the solution to yielding perfect edges. Noise, obstructions, sharpness, exposure, empirical parameters, and more all affect the final edge quality. The WVF and LF already handle noise and obstructions very well due to their unique architecture and have very complementary attributes to handle the edge capture of small objects and edges with small discontinuities. However, when used on their own, they have limitations. The WVF is capable of handling complex shapes but does not perform as well when presented with high noise of line breakages. The LF on the other hand is less adept regarding capturing complex shapes. It is therefore natural to combine the two in a scale space, which allows them to handle all the above and better replicate the human vision system. To build the scale space, the filters are run independently and then combined into one gradient map using a Gaussian Mixture Model (GMM). This method allows any number of filters to be run and combined and allows the input of multiple sensor domains for the same image.

Single domain edge detection is limited in that it can only consider the information from one physical spectrum. Edges produced from a monochrome image are high quality since they use no color filters and, therefore, capture with a higher signal to noise ratio as compared to color. On the other hand, edges from color images have an added layer of error introduced from Bayer filtering but do enhance information in each color domain. Both domains, however, could have poor representation in the low-light condition or contain significant noise which degrades small objects or those with smooth intensity boundaries. A rope or hose in water may resemble a wave crest in a monochrome image but will stand out with color information while a buoy at night may not be visible to color or monochrome but can be detected in IR and Thermal imagery. The GMM allows for the inclusion of all such image domains for a more comprehensive gradient analysis where the high signal ratio of monochrome, the RGB information from color, and other added sensor modalities are all preserved in the final edge map.

The scale and domain space are built by combining the gradients at a pixel according to their most prominent distribution with respect to their orientation. This is accomplished by creating a set of gradients from various numerical derivative and smoothing filters. All the gradients are stacked on top of each other to create a tensor, where each pixel contains a variety of gradients with respective orientations. It can be used to establish a probability density function, where the largest gradient magnitude corresponds to the dominant direction and all other gradients correspond to noise. This GMM not only aids in creation of smooth connected gradients for similar edges but allows the introduction of additional image domains, such as monochrome, RGB, IR, and others. This allows the gradients in the image to avoid a lot of issues present with varied lighting conditions, poor exposure, low contrast, and more.

The process for determining high quality edges beings with computing gradients from the WVF and LF and immediately follows with determining their orientations.

A. Computing Gradient Orientaion Using Cubic Splines

Gradient orientations serve as a valuable piece of information in many vision techniques and are critical for edge detection with non-maximum suppression. Traditionally, edge orientations are computed using the x and y gradients obtained from directional operators. They are obtained using the arctan of the calculated x and y gradients. This is prone to errors if the x and y gradients are unreliable due to noise, low contrast, or subpixel edge perturbations [23]. Additionally, the discrete nature of the image domain combined with the small size of commonly accepted gradient detection kernels introduces significant inaccuracies. Sampling from a discreet domain means the data used is limited to a specific set of values and corresponds to a certain resolution. This limits the set of angles that can be perfectly represented by an image and subsequently captured in a kernel. When considering this along with the 90-degree separation of x and y gradients, it is nearly impossible to estimate an edge's angle with any degree of acceptable error.

The proposed method leverages a unique gradient operator, the WVF and LF, to produce gradients at many angles, limited only by computation, and locate the largest magnitude with periodic cubic splines. For more details of WVF and LF, please refer to [21]. All the derivative values produced by the WVF and LF can then be used as data points along with their respective angle values to form a smooth function describing the relationship between filter orientation and gradient value. These data points are used to fit a cubic periodic spline, which establishes the relationship between the derivative values and the orientation angles. Cubic periodic splines can fit piecewise functions to bridge data points which are smooth about the first and second derivative. This results in a smooth flowing function describing an entire set of data. Since an edge will occur where the gradient value is largest, we can locate the maximum along this spline and determine the corresponding angle. This can be done by finding the maximum value in the cubic spline or the point in which its slope is zero. This vastly enhances angular accuracy (visualized in Figure 1).

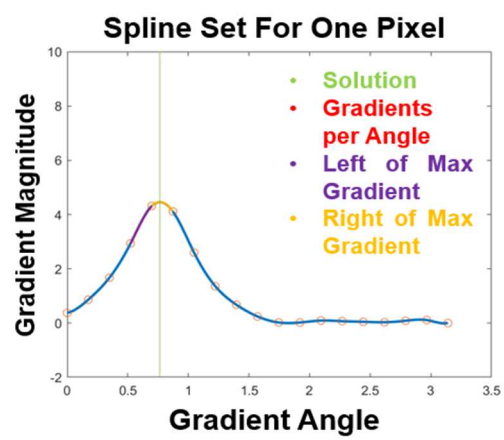

Figure 1: Solving Gradient Orientation at Spline Maximum for a Pixel

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B. Angle Detection Process

The LF is first run at a user specified number of angles, defining the angular resolution. The derivative maps produced by the LF at this angular resolution are fed in as the input of the angle detection script. Each pixel will have multiple derivative values calculated at various orientations by the line filter. These derivative values along with the orientation of the line filter that produced provide the data point to build the periodic cubic spline at the indexed pixel.

The spline provides a set of piecewise function continuous to the second derivative (Figure 1). This can be leveraged to find the angle value, which produced the maximum derivative value for the pixel. First, the maximum derivative that the line filter was able to produce is found and then indexed on the spline. The maximum gradient from all angles processed by the filter determines which splines must be analyzed to find the global maximum across the spline function. The global maximum will be found on one of the splines to the left and right, so these two splines will be analyzed. The piecewise functions that precede this value and come after are examined for zero slope values along their respective intervals. Zero slope locations correspond to local minimums and maximums on the spline. Only one zero derivative value should exist for the two functions when analyzed over their respective domains.

The angle location at which the derivative's zero value occurs is defined as the angle of the gradient as that is where the maximum derivative occurs. In cases where the indexed pixel lies on an edge, the derivative values will be large and easy to process. However, in cases where the pixel lies in noise or in the background, the derivative values will be much smaller and fluctuate dramatically with the orientation angle. By measuring the range of the derivative values produced on a pixel's spline and comparing them to the overall range of derivatives found in the image, the indexed pixel can be disregarded if its derivative range is too small.

After the angle is calculated for the indexed pixel, the script moves to the next pixel in the image and repeats the process to calculate its angle. All the calculated angles are then stored in a two-dimensional matrix which will be used to create a histogram displaying the frequency of all the angles calculated in the image. This histogram represents the frequency of angles in the image and can be used for further processing to group edges based on their angles.

Once the matrix of accurate angles is obtained, it can be used for better non-maximum suppression (NMS) of gradients in Canny like edge detection processes. However, normal NMS is susceptible to breakages at corners and areas of high curvature. [24] For this reason, we have added on to the typical NMS process. In this methodology, the angular fidelity of NMS is doubled, so in addition to the directions of 0, 45, 90, and 135 degrees, we also examine 0, 22.5, 45, 67.5, 90, 112.5, 135, and 157.5. This helps reduce breakages on edges with complex geometry. Additionally, both the maximum gradient at a pixel and the second maximum gradient are used for comparison. This helps areas like corners, which have two dominant gradients. Typically, only one of the directions will be used when thinning the gradient, however we will double check the second maximum to see if it is comparably large to the maximum gradient and then both directions will be preserved.

C. Gaussian Mixture Model for Building Gradient Scale Space

Using the Gaussian mixed model for edge determination is simple in principle. Multiple channels can supplement each other to provide edge information where another channel may be lacking. To start, each channel will be individually processed with the multi-scale LF for gradient calculation. This results in a set of gradient maps for each channel, which correspond to the gradients produced from each size of the filter used to process each respective channel. This describes each pixel with multiple gradients, which will allow us to determine which channel and filter produce the largest gradient. Depending on the filters and channels used, this reveals information about the line likeness of the object associated with the pixel as well as its local intensity and color gradient.

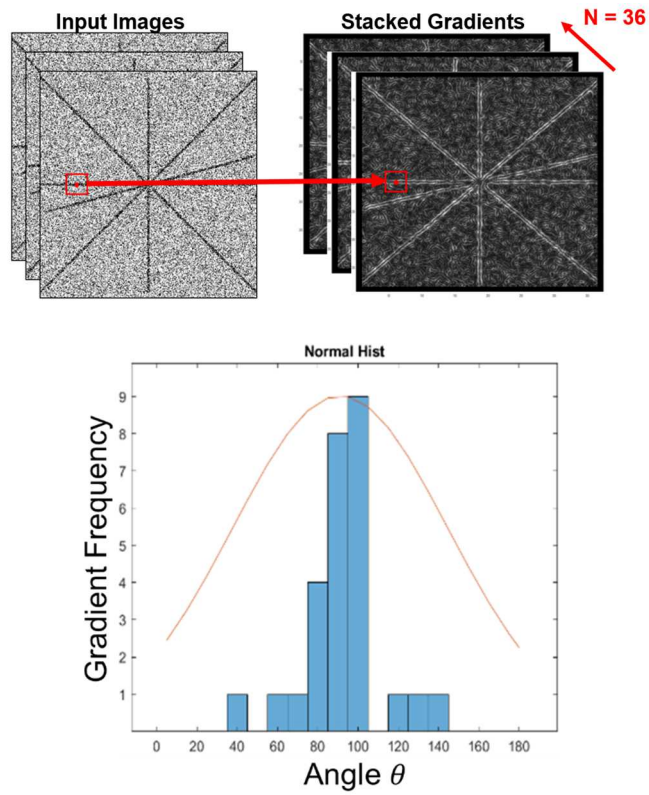

Figure 2: Probability Density Function for 90 Degree Line Pixel and the Associated Input Images and Gradient Stack

Once all individual gradient maps are obtained, they can be stacked together to build a 3-D array where the first two dimensions correspond to the x and y pixel location, and the third dimension corresponds to gradients at each channel and filter size. Given enough information in the third dimension, a probability distribution can be constructed to describe the gradient at each pixel in terms of edge vs. noise gradients with their associated orientation and magnitude. For example, if 36 gradient maps can be constructed from various filters and

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channels, each pixel can generate a probability density function (PDF) containing its associated 36 gradient values binned according to their orientation. This PDF would display how many gradients were collected at that pixel location for each orientation bin. Figure 2 contains a simple example showing how a PDF can be generated at a pixel in this scenario. A pixel is indexed along the 90-degree line in the image, and its gradients are collected from the multiple channels provided.

Intuitively, we would expect to see a concentration of gradients around the 90-degree bin in this distribution. We do see exactly this behavior; however, we also see outlier gradients dispersed throughout the histogram which are associated with noise. We know that edges produce strong gradients at specific orientations, and we also know that noise is random and will not produce gradients at the same orientation consistently. This can be leveraged to suppress the noise in the histogram and pronounce the true edge gradients, which is proven when examining the gradient magnitude weighted histogram in Figure 3. The process of weighing the histogram is straightforward. Instead of counting the number of gradients at each angle to determine the height of the bar, the gradient values within each bin are summed together. This means noise gradients which produce small values will be minimized in the histogram.

Once a gradient weighted PDF is obtained, we can begin to decompose the distribution to selectively obtain the bins and pixels which correspond to true edges while discarding any remaining noise contributions. Gaussian mixture modeling is a commonly used technique in computer vision and data analysis, which works off the assumption that any distribution can be broken down into multiple smaller gaussian distributions. In the case of edge determination, we know that our gradient distributions will consist of two components, edge distributions and noise distributions. The edge distributions can be found by selecting the distribution with the largest mixing coefficient since it will dominate the weighted PDF. This concept is visualized in Figure 3 and is achieved using the expectation maximization (EM) algorithm. The EM algorithm is an iterative process, which works in a two-step guess and check process to maximize likelihood of estimates for statistical parameters. In this case, the distribution is broken down into two Gaussians which are located around the same mean of 90 degrees where the edge PDF has a very prominent peak and small standard distribution and the noise PDF has a much larger standard distribution and small peak.

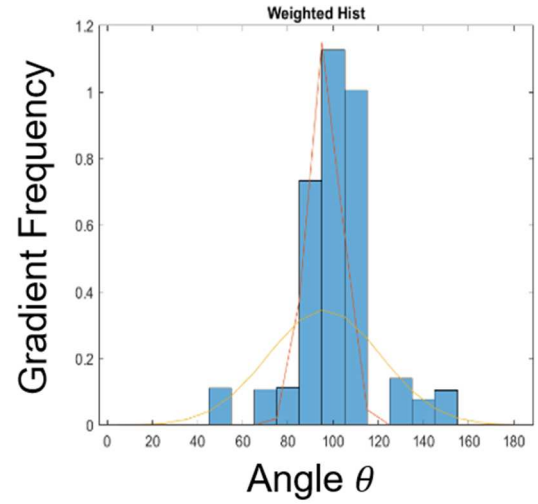

Figure 3: Weighted Histogram with GMM Decomposition Overlaid

Once the edge distribution is found, it can be used to estimate the layered multichannel gradient at the pixel location along with its associated orientation. This process is repeated for every pixel in the image and is expandable to include as many channels as remains sensible. After applying the GMM to the stacked gradients, the output will be one gradient map which encompasses all input layers.

IV. Results and Analysis

Because it is challenging to quantitatively evaluate the performance of edge detection these results are divided into three portions. First the angles generated from cubic splines are compared against angles from the arctan method. This is done using a grouping test on a synthetic image with lines of known orientation. This allows us to examine the consistency of angle prediction under noisy conditions. After, edges are generated with each of the angles using a photo from a popular edge detection dataset BIPED. They are visualized next to each other to examine the differences, and a difference map of the two is provided to show the superior connectivity from cubic spline angles.

Next, a few images from a marine environment are used to reveal the effects of using a scale space and multiple image domains with the WVF and LF filters. Since ground truth (GT) for edge detection only comes from human labeled data or data produced by a subjective benchmark the first analysis is qualitative. Qualitatively, the most important evaluation criteria are gradient smoothness, edge gradient contrast, angular invariance, noise suppression, and edge connectivity. All these metrics can be assessed visually.

Afterwards the filters were applied to three different datasets to generate numerical evaluation metrics against state-of-the-art (SOTA) machine learning models. The datasets used are UDED [20] and BIPED [17], which are very popular in evaluating edge detection methods because of their labels. Our method was compared against DexiNed and TEED for generating these results. This allows for a quantitative analysis of the methods when presented with real world scenarios

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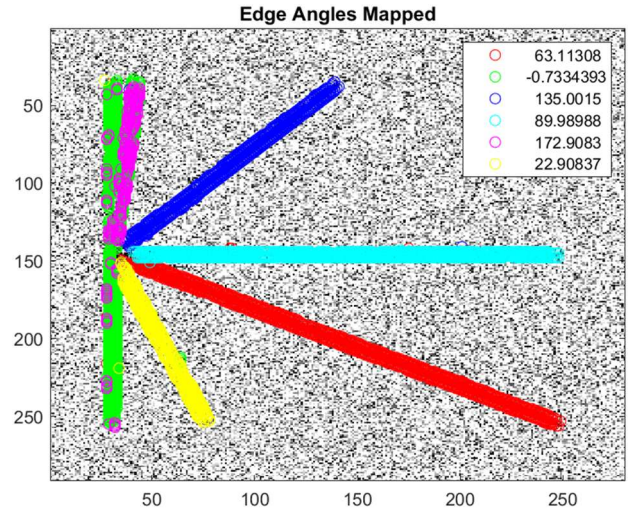
containing complex objects. When generating the ODS, OIS, and mAP edges were forward and backward matched within a radius of 3 pixels to account for ambiguity in the ground truth labels, which is unavoidable. Another common method to address this is to include a multiscale edge match. However, this was not included, as it appears to depress the scores.

To ensure that fair standards are set when comparing our method to the machine learning models, we outlined a process for determining edges with thresholds, which is equivalent across the models. Canny outlines a highly regarded process used to obtain edges from filter gradients [6] and as such, gradients and probabilities from each operator are paired with the Canny determination process for obtaining edges in all results. The thresholds tested ranged from 0-1 and incremented by 0.001 every iteration meaning a total of 1001 thresholds were used for each photo.

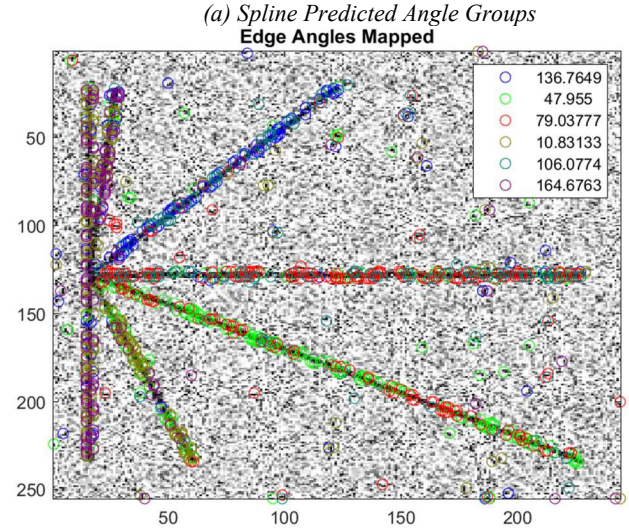
A. Angle Detection Results

To prove how computing gradient angles from cubic splines is much more accurate than using an arctan decomposition, two tests were carried out. First, angles were generated for a synthetic image with multiple lines at various angles with a signal to noise ratio (SNR) of 2. While an SNR of 2 is noticeable, it is not unreasonable to see in a real image under the right conditions and is a good test. After generating the angles at every pixel, the noise was discarded based on a light thresholding. One advantage of the cubic spline method is that noise can be more intelligently discarded by analyzing the range of gradient values at a pixel according to the number of gradients that were produced by the rotated filters. A small range indicates the pixel is noisy because noise should not produce salient gradient values in any direction. This is not possible when doing a simple xy decomposition, and the result is therefore noisier. Once the noise was discarded, the angles were grouped according to their density and then displayed on the photo with their calculated angle value based on the centroid of the group. These results are shown in Figure 4.

It can be easily seen from the images, the angles produced using cubic splines are vastly superior to those of the arctan decomposition. It can immediately be seen that the method based off arctan has much more noise, and the angle predictions are not smooth across line segments. Each line segment in this example contains various angle predictions. The cubic spline method is instead smooth and accurate with minimal noise. It only runs into small inaccuracies when lines are closely oriented. Table 1 shows the numerical values predicted for each line versus their true values. The cubic spline consistently predicts angles within one degree of their true value while the arctan method is all over the place.



(a) Spline Predicted Angle Groups



(b) Arctan Predicted Angle Groups

Figure 4: Angle Group Predictions using Cubic Spline Method and Arctan Method

Table 1: Predicted Angle Values for Synthetic Image

True Angle	Spline Predicted Angle	Arctan Predicted Angle
0	-0.733	10.831 and 164.676
23	22.908	10.831 and 47.955
63.5	63.113	47.955 and 79.037
90	89.990	79.037 and 106.077
135	135.002	136.765
174	172.908	164.676 and 10.831

The second test executed for evaluating the cubic spline angle detection was done by generating edges for a real image from the BIPED dataset using cubic spline angles and arctan angles. Since angles are crucial to non-maximum suppression, any error in the angle predictions can significantly degrade the results. Qualitatively, the effects can be observed from the results in Figure 5. The edges produced by the arctan method are very broken, and a high amount of noise exists. On the contrary, the edges produced with the cubic spline method are smooth, connected, and have much less noise. This is

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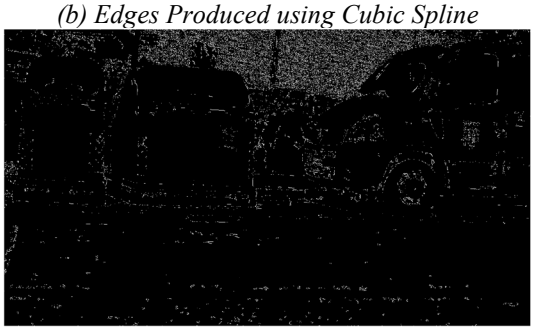
particularly obvious when examining complex shapes like the chain link above the car or the license plate.

(a) Edges Produced using ArcTan

(b) Edges Produced using Cubic Spline



(c) Difference Map

Figure 5: Comparison of Edges Generated with Arctan Angles and Cubic Spline Angles

B. Gaussian Mixture Model Multiscale Edge Results

Two images were processed to demonstrate the effects of building a scale space in edge detection as well as using multiple input domains. The first scene was captured in monochrome under less-than-ideal lighting conditions. It displays an underexposed scene from the POV of a boat looking at two buoys, a crab pot, and a speedboat. The overlayed edges can be seen in Figure 6. It is clear that when creating edges using only the WVF, the objects themselves are captured. However, there is a large amount of noise and breakages on the lines. Using only the LF, the edges are very smooth and connected, but curved lines and complex shapes cannot be captured. When combining the two, the best attributes of each are preserved. The combined edges ignore most of the image noise, capture all objects and avoid most of the discontinuities.

Next, the effects of including multiple image domains were analyzed. A photo containing a braided cable floating among a small wave field was used because it provides an exceptional challenge for edge detection. The cable itself is floating on the surface but is submerged under the moving water in places, this means it is extremely hard to capture smooth connected edges. Additionally, the harsh water texture acts as surrounding noise which degrades the edges in multiple locations. This is especially true when observing the monochrome layer only as it blends in very well with the water. Once color is introduced, it does stand out, but all the listed effects of the water are still true. After color, hue is introduced to provide an additional layer of contrast for the hose and suppress some of the water texture. This improves the final edge map very well. These results are visible in Figure 7.

C. Numerical Analysis

To ensure the performance of this edge detection method is not simply qualitative, three datasets were used to generate evaluation metrics against two SOTA machine learning edge detection models. Specifically, our method is compared against DexiNed and TEED. DexiNed is a full-scale model, which takes a little longer to train and has more parameters while TEED is a compact and efficient model with comparable performance. The metrics used for comparison are the optimal dataset Scale (ODS), Optimal Image Scale (OIS), and mean average precision (mAP). These are commonly used metrics for analyzing performance over a dataset.

Two common datasets, UDED and BIPED were chosen for generating these metrics because they were hand crafted for edge detection analysis. UDED combines some of the more

Figure 6: Comparisons of Scale Space Edges Vs. Single Scale Edges

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Figure 7: Comparison of Single Scales and Single Domains Versus Multiscale Multidomain Edges

challenging images from a variety of datasets to ensure the images are challenging. BIPED is a dataset which has been hand labeled by a group of researchers and edges are determined by the aggregate score from those labels. These two are popular benchmarks and the results of processing images from them with our method versus DexiNed and TEED are presented below in Table 2. In this table, it can be seen our method outperforms both machine learning models in every metric when looking at the UDED dataset, indicated by the scores in green. When looking at the BIPED dataset, however, our model only wins in terms of mAP. It does still generate good ODS and OIS scores but is slightly edged out by TEED. This is expected as both TEED and UDED were trained on the BIPED dataset. While the test data used to generate these scores was separate from the training data, it is still similar and may have been more identifiable for the models.

Table 2: Numerical Results for BIPED and UDED Datasets

Method	EPOCH	ODS	OIS	mAP
DexiNed-UDED	11	.5541	.5516	.5587
DexiNed-BIPED	11	.6584	.6593	.5184
TEED-UDED	6	.6239	.6253	.6474
TEED-BIPED	6	.6976	.6980	.6528
WVF/LF MultiScale (Ours) – UDED	N/A	.7185	.7274	.7680
WVF/LF MultiScale (Ours) – BIPED	N/A	.6594	.6675	.7068

Since this edge detection method was designed to perform well in aquatic environments, especially when presented with small objects and challenging lighting conditions, a very small dataset was generated and hand labeled to generate the same metrics with. This dataset contains four images, all looking at water, with a lot of noise, harsh lighting, and poor exposure. The results using our method versus DexiNed and TEED on this data are presented in Table 3. While our method has a slightly lower ODS score than DexiNed in this test, it vastly outperforms both methods in every other metric as indicated by the highest scores outlined in blue.

Table 3: Numerical Results for Aquatic Data

Method	EPOCH	ODS	OIS	mAP
DexiNed - Aquatic	11	.6669	.6551	.5447
TEED - Aquatic	6	.5692	.6242	.5741
WVF/LF MultiScale (Ours) – Aquatic	N/A	.6284	.7411	.7795

V. Conclusion

The two proposed methods, including spline-based angle computation and multi-scale, multi-domain edge determination vastly improve edge detection performance. The angle calculation can predict pixel orientations to within a degree, and the improved edge determination method produces highly localized, smooth, and continuous edges resulting in increased true positive rate, recall, precision, and f-score while decreasing false positive rate. When compared to SOTA learning-based models the multi-scale and multi-domain edge detection method can outperform, especially when presented with challenging images. This is especially true when looking at aquatic images, which is proven by the ODS, OIS, and mAP scores presented.

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